

Options within Variable Annuities

11.1 To Live and Die in VA

In this chapter I will focus on exotic options and derivative securities that are embedded within tax-sheltered saving policies called *deferred* variable annuities, which are distinct from the immediate variable annuities discussed in previous chapters. In the United States, a variable annuity (VA) contract is similar to an open-ended mutual fund (“unit trust” in the United Kingdom) except that all investment gains are tax-sheltered until the money is withdrawn or annuitized. Moreover, a variable annuity has a unique form of investment protection: in the event of death, the variable annuity will pay out the greater of the account market value and the original investment grown at a fixed rate. More recent versions of variable annuities have additional guarantees—such as living benefits—that I will discuss a bit later.

Call and put options are the building blocks of most derivative securities and thus are at the heart of most structured products in finance. As described in Chapter 9, a call option provides the holder with the right but not the obligation to purchase (or call) an underlying security at some fixed price (the strike price). If we let S_t denote the price of the underlying security at time t , then a call option pays off

$$\max[S_t - X, 0],$$

where X is the strike price. A put option gives the holder the right but not the obligation to sell (or put) the underlying security at some fixed price. The put option pays off

$$\max[X - S_t, 0],$$

where again X is the strike (or sale) price and S_t is the *spot price* of the underlying security at time t . An American option can be exercised at any

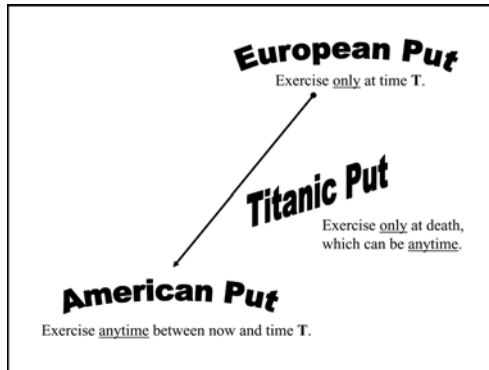


Figure 11.1. Three types of puts

time $t \leq T$, where T is the maturity, whereas a European option can be exercised only at maturity $t = T$. Obviously, the value of an American option should be at least as much as the value of a European option since the rights of the latter are contained within the former. See Figure 11.1.

I label the options that mature at a random time of death “Titanic” options because they are somewhere between American- and European-style options. A famous result in financial economics is the Black–Scholes/Merton (BSM) valuation formula for the price of a European put and call option, which can be written as

$$\text{BSP}(T) = Xe^{-rT}\Phi(-d_2) - S_0e^{-kT}\Phi(-d_1) \quad (11.1)$$

and

$$\text{BSC}(T) = S_0e^{-kT}\Phi(d_1) - Xe^{-rT}\Phi(d_2), \quad (11.2)$$

respectively. Here $\Phi(z)$ denotes the standard normal cumulative distribution function (or the “area under the curve” from negative infinity to z), k denotes any dividends (expressed as a yield) that are paid on the underlying security, and d_1 and d_2 are defined by

$$d_1 = \frac{\ln[S_0/X] + (r - k + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (11.3)$$

Often the strike price X is set equal to the original security spot price S_0 , which is then arbitrarily set to $S_0 = 1$, so the $\ln[S_0/X]$ term become zero. In other cases the strike price increases over time at a (guaranteed) rate $g \geq 0$. If $X = e^{gT}$ then the put formula can be simplified to

$$\text{BSP}(T) = e^{(g-r)T} \Phi(-d_2) - e^{-kT} \Phi(-d_1) \quad (11.4)$$

with

$$d_1 = \frac{(r - k - g + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \quad (11.5)$$

in which case the put option value will be expressed as a percentage of the underlying security.

Either way, the formula is based on the assumption that the underlying security price “obeys” or satisfies the equations of (Brownian) motion specified in Chapter 5, namely, that

$$\frac{\Delta S_{i+1}}{S_i} = \left(v + \frac{1}{2}\sigma^2 - k \right) \Delta t + \sigma \Delta N(0, \sqrt{\Delta t}). \quad (11.6)$$

This implies that the proportional change in the value of the state variable is equal to the sum of two terms. The first term is a deterministic increase of $\mu := v + \frac{1}{2}\sigma^2$ minus a dividend yield of k , multiplied by the incremental change in time Δt . The second term is the random component. In the continuous-time limit, (11.6) converges to the following stochastic differential equation (SDE):

$$\begin{aligned} \frac{dS_t}{S_t} &= \left(v + \frac{1}{2}\sigma^2 - k \right) dt + \sigma dB_t \\ &= (\mu - k)dt + \sigma dB_t. \end{aligned} \quad (11.7)$$

I have elaborated on the intuition behind this equation in Chapter 4. Table 11.1 provides some numerical examples for the BSM (put) formula under a variety of maturity values t .

A quick-and-dirty explanation for “where” the valuation formula comes from can be obtained by going through the calculus:

$$\text{BSM}(t) = \int_{-\infty}^{\infty} \max[X - S_0 e^{\tilde{g}T}, 0] d\tilde{g}, \quad (11.8)$$

where $\tilde{g}$ is the normally distributed “annualized growth rate” (but assuming that $E[\tilde{g}] = r + \frac{1}{2}\sigma^2$ instead of the usual $E[\tilde{g}] = \mu + \frac{1}{2}\sigma^2$) and $\text{SD}[\tilde{g}] = \sigma/\sqrt{T}$. In terms of notation, I will write $E^*[\tilde{g}] = r + \frac{1}{2}\sigma^2$ when the expectation uses $\mu = r$ and $E[\tilde{g}] = \mu + \frac{1}{2}\sigma^2$ when the expectation uses the standard μ . Remember, yet again, that we replace the arithmetic mean μ with r .

Table 11.1. *BSM put option value as a function of spot price and maturity—Strike price = \$100*

Spot price	Maturity <i>t</i> (years)						
	1	2	3	4	5	6	7
\$10	\$84.28	\$78.89	\$73.82	\$69.05	\$64.57	\$60.35	\$56.39
\$20	\$74.38	\$69.09	\$64.12	\$59.46	\$55.11	\$51.06	\$47.30
\$30	\$64.47	\$59.29	\$54.47	\$50.03	\$45.99	\$42.33	\$39.00
\$40	\$54.58	\$49.56	\$45.08	\$41.15	\$37.68	\$34.61	\$31.85
\$50	\$44.70	\$40.13	\$36.36	\$33.20	\$30.47	\$28.07	\$25.92
\$60	\$35.01	\$31.41	\$28.67	\$26.41	\$24.43	\$22.68	\$21.09
\$70	\$25.94	\$23.81	\$22.20	\$20.79	\$19.51	\$18.31	\$17.19
\$80	\$18.08	\$17.56	\$16.95	\$16.27	\$15.54	\$14.80	\$14.06
\$90	\$11.87	\$12.65	\$12.81	\$12.68	\$12.38	\$11.99	\$11.54
\$100	\$7.38	\$8.95	\$9.62	\$9.87	\$9.88	\$9.74	\$9.51
\$110	\$4.38	\$6.24	\$7.19	\$7.68	\$7.90	\$7.94	\$7.87
\$120	\$2.50	\$4.31	\$5.36	\$5.98	\$6.33	\$6.50	\$6.55
\$130	\$1.39	\$2.96	\$3.99	\$4.66	\$5.09	\$5.34	\$5.47
\$140	\$0.75	\$2.02	\$2.97	\$3.65	\$4.10	\$4.40	\$4.58
\$150	\$0.40	\$1.37	\$2.22	\$2.86	\$3.32	\$3.64	\$3.86
\$160	\$0.21	\$0.93	\$1.66	\$2.25	\$2.70	\$3.03	\$3.26
\$170	\$0.11	\$0.63	\$1.24	\$1.77	\$2.20	\$2.53	\$2.77
\$180	\$0.05	\$0.43	\$0.93	\$1.40	\$1.80	\$2.11	\$2.36
\$190	\$0.03	\$0.29	\$0.70	\$1.11	\$1.48	\$1.78	\$2.01
\$200	\$0.01	\$0.20	\$0.53	\$0.89	\$1.22	\$1.50	\$1.73

Note: Risk-free rate = 6%, dividend yield = 1%, volatility = 25%.

11.2 The Value of Paying by Installments

One of the many unique aspects of derivative securities that are embedded within life and pension annuity products is that—in contrast to over-the-counter and exchange-traded financial derivatives, where the option premium is paid up front—the payment for these derivatives is made in installments. These installments are often structured as an asset-based fee that is proportional to the account value itself. In other words, you don’t really know what you will end up paying (and the company takes the risk of not knowing what fees they will be receiving). Thus, for example, you might pay 1% of the market value of the asset, charged and withdrawn monthly by multiplying the account value by 0.01/12. I stress that this is quite different from what happens when buying a generic call or put option, where the premium (of, say, 5% or 10% of the notional value) is paid as soon as the contract is initiated.

My objective in this section is to value the ongoing stream of proportional insurance/derivative fees and then discount this cash flow to time 0 so I can eventually compare the present value of the benefits you receive from these various guarantees to the present value of the costs that you pay for them. As in the rest of the book, I will discount these fees assuming they are paid and deducted from the account in continuous time until some deterministic or stochastic maturity horizon. At a crude level I am trying to address the following question: What is the discounted value of the 1% or 2% insurance fee you will be paying on your investment account—which is currently worth \$10,000, for example—during the next 5, 10, or 20 years? On the one hand, given that we do not know exactly how this investment account will perform over time, it should be difficult to value 1% or 2% of a random stream. However—and here is the counterintuitive aspect of this exercise—we can actually obtain a present value by “self-replication” arguments. Moreover, the risk-neutral present value does not depend on interest rates.

I compute the discounted value of the insurance/derivative risk charge by treating the stochastic cash flows as a contingent claim on the underlying account value. The insurance company can be viewed as having a long position in the continuous-flow fee derivative. The derivative remains alive as long as the policyholder has not died or lapsed the policy. Therefore, we can model the general risk-neutral evolution of the underlying account or asset price via the usual

$$dS_t = (r_t - k_t)S_t dt + \sigma(S_t, t)S_t dB_t, \quad S_0 = 1, \quad (11.9)$$

where B_t is the by-now familiar Brownian motion, r_t is a (possibly stochastic) interest rate (but assumed independent of S_t), k_t is the insurance/derivative fee, and $\sigma(S_t, t)$ represents the (possibly stochastic) volatility of the underlying security. The integral representation of equation (11.9) is:

$$S_t = S_0 + \int_0^t (r_u - k_u)S_u du + \int_0^t \sigma(S_u, u)S_u dB_u. \quad (11.10)$$

In the special case of geometric Brownian motion—which was the core of the presentation in Chapter 5—the volatility parameter $\sigma = \sigma(S, t)$ is constant.

I now need to define a new “money market” investment account, which is denoted by

$$R_t = \exp\left\{\int_0^t r_s ds\right\} \quad (11.11)$$

and can be viewed as the future value of one dollar that is invested at a rate of r_t . When this rate is constant the future value factor will be $R_t = e^{rt}$, but when the rate is stochastic the future value will be random. The interesting result—which I will soon prove—is that, regardless of how complicated one makes the interest rate dynamics for r_t , the expected (risk-neutral) discounted value of the proportional fees k_t will be identical!

To show this, let K_t denote the stochastic value (discounted to time 0) of insurance/derivative fees collected until time t . By construction, we have

$$dK_t = R_t^{-1} k_t S_t dt. \quad (11.12)$$

The quantity $k_t S_t dt$ can be viewed as the instantaneous earnings of the insurance company that is charging the proportional fee of k_t , while the R_t^{-1} factor discounts the quantity to time 0. Our main objective now is to obtain values both for K_τ and its expectation $E[K_\tau]$, where τ is a *general* stopping time for the process S_t . In English: What is the present value of paying 2% of the account value from now (time 0) until some later time τ ?

By a simple chain rule, we have

$$\begin{aligned} d(R_t^{-1} S_t) &= -r_t R_t^{-1} S_t dt + R_t^{-1} dS_t \\ &= -r_t R_t^{-1} S_t dt + R_t^{-1} (r_t - k_t) S_t dt + R_t^{-1} \sigma(S_t, t) S_t dB_t \\ &= -R_t^{-1} k_t S_t dt + R_t^{-1} \sigma(S_t, t) S_t dB_t \\ &= -dK_t + R_t^{-1} \sigma(S_t, t) S_t dB_t. \end{aligned} \quad (11.13)$$

Therefore, by re-arranging equation (11.13) and noting that (by definition) $R_0^{-1} S_0 = 1$, we obtain

$$\begin{aligned} K_\tau &= \int_0^\tau dK_t = - \int_0^\tau d(R_t^{-1} S_t) + \int_0^\tau R_t^{-1} \sigma(S_t, t) S_t dB_t \\ &= 1 - R_\tau^{-1} S_\tau + \int_0^\tau R_t^{-1} \sigma(S_t, t) S_t dB_t. \end{aligned} \quad (11.14)$$

So the discounted value of the insurance/derivative fee up to a stopping time τ is $1 - R_\tau^{-1} S_\tau$ plus an integral term whose expectation is zero. This implies that

$$E[K_\tau] = 1 - E[R_\tau^{-1} S_\tau]. \quad (11.15)$$

In specific cases, equation (11.14) can be solved to provide the entire distribution of the discounted value of fees. More importantly, equation (11.15) can be easily applied to a variety of stochastic maturities. Here are some examples of different maturities.

If τ is *deterministic*, $r_t = r$, $\sigma(S_t, t) = \sigma$, and $k_t = k$, then the stochastic differential equation in (11.9) can be solved to yield

$$S_t = \exp\left\{\left(r - k - \frac{1}{2}\sigma^2\right)\tau + \sigma B_\tau\right\}; \quad (11.16)$$

hence (11.15) can be simplified to

$$E[K_\tau] = 1 - E\left[\exp\left\{\left(-k - \frac{1}{2}\sigma^2\right)\tau + \sigma B_\tau\right\}\right] = 1 - e^{-k\tau}. \quad (11.17)$$

Notice that the interest rate r and the volatility σ drop out of equation (11.17), so that the risk-neutral expected discounted value of fees is invariant with respect to both parameters. For example, an insurance risk charge of $k = 0.02$ (2%) with $\tau = 20$ yields $E[K_{20}] = 0.329$, which implies that an investor with a 20-year horizon is implicitly paying 33% of the initial account value. In contrast, if $k = 0.002$ (20 basis points) then $E[K_{20}] = 0.039$, which is less than 4% of the initial account value.

If τ is *stochastic* but independent of S_t , equation (11.15) leads to expectations with respect to both random variables. In many cases $\tau = T_x$, where T_x is the remaining lifetime random variable with probability density function $f_x(t)$. Once again, with $r_t = r$, $\sigma(S_t, t) = \sigma$, and $k_t = k$, we condition on age x to obtain

$$\begin{aligned} E_x[K_{T_x}] &= 1 - E_x\left[\exp\left\{\left(-k - \frac{1}{2}\sigma^2\right)T_x + \sigma B_{T_x}\right\}\right] \\ &= 1 - E_x\left[E\left[\exp\left\{\left(-k - \frac{1}{2}\sigma^2\right)T_x + \sigma B_{T_x}\right\} \mid T = t\right]\right] \\ &= 1 - E_x[e^{-kT}] = 1 - \int_0^\infty e^{-kt} f_x(t) dt, \end{aligned} \quad (11.18)$$

which some readers might identify as (1 minus) the Laplace transform of the remaining lifetime random variable evaluated at k . In fact, when $f_x(t) = \lambda e^{-\lambda t}$, which is the exponential remaining lifetime, (11.18) leads to

$$E_\lambda[K_T] = 1 - \lambda \int_0^\infty e^{-(k+\lambda)t} dt = \frac{k}{\lambda + k}. \quad (11.19)$$

The λ subscript replaces the current age (x) as the “conditioning” variable. For a 65-year-old with an expected future lifetime of $E_\lambda[T] = \lambda^{-1} = 20$ years, using $k = 0.02$ as a proportional asset-based insurance/derivative fee leads to a present value of $E_\lambda[K_T] = 0.2857$, or about 28% of the initial account value. But if $k = 0.002$ then $E_\lambda[K_T] = 0.038$, which is less than 4% of the account value. Notice that both values are strictly lower than (a naïve application of) $1 - e^{k/\lambda}$. This is a consequence of Jensen’s inequality, according to which $1 - E_\lambda[e^{-kT}] < 1 - e^{kE_\lambda[T]}$.

Table 11.2. *Discounted value of fees*

Charge k	Time (years)		
	10	20	30
25 b.p.	2.47%	4.87%	7.22%
50 b.p.	4.87%	9.51%	13.92%
150 b.p.	13.92%	25.92%	36.23%

Notes: b.p. = basis points. Interest rates, expected returns, and volatility are irrelevant.

Finally, under the more realistic GoMa specification of the function $f_x(t)$, by (11.18) we have

$$\begin{aligned} E_x[K_T] &= 1 - \int_0^\infty e^{-kt} f_x(t) dt \\ &= kb\Gamma(-bk, b\lambda(x)e^{-(m-x)k+b\lambda(x)}), \end{aligned} \tag{11.20}$$

where $\lambda(x)$ = GoMa force of mortality at age x and $\Gamma(a, b)$ is the incomplete Gamma function (see Chapter 3). The quantity $1 - E_x[K_T]$ can also be identified as the net single premium for a life insurance policy under a force of interest k and future lifetime density $f_x(t)$.

The reader should now have a collection of formulas that can be used to compute the (risk-neutral) present value of a random series of cash flows that result from charging $k\%$ of the account value in continuous time. This is what you pay for living and death benefits.

The key insight, once again, is that equation (11.17) does not contain any mention of the interest rate in the market. This is most counterintuitive. Normally one expects that some sort of interest rate or term structure is needed to arrive at the present discounted value of cash flows. However, when these cash flows are stated in proportional terms of an account value, the discounted value is no longer a function of the interest rate.

Table 11.2 should make this point clearly; it displays the result of paying proportional fees of 25, 50, and 150 basis points on an investment account for the next 10, 20, and 30 years. According to the table, if you plan to invest \$100 in a variable annuity (or in any investment account, for that matter) for the next 20 years and if the company holding the funds charges 50 basis points per year, then the discounted value of this fee at time 0 is 9.51% of the current value of the account, or \$9.51. If the fee charged is 150 basis points,

then the discounted value of this fee during the next 20 years is 25.92% of the account.

Some readers might wonder how I can make this assertion without any consideration of interest rates or projected growth rates for the account. This is a legitimate concern, and let me offer the following numerical example to explain. Imagine that Mr. Pay invests \$100 for the next 20 years in a portfolio that is charging 150 basis points per year while Mrs. Free invests \$74.08 in an identical investment portfolio that does not charge any management fee. Mrs. Free starts off with much less than Mr. Pay, yet both are invested in the same underlying financial instruments. After 20 years, Mrs. Free's portfolio grows to a random $74.08e^{g_{20}}$, while Mr. Pay's portfolio grows to a random $100e^{g_{20} - (0.015)20}$, where the extra $(0.015)20$ in the exponent denotes the 150 basis points subtracted in management fees. Observe that $100e^{-(0.015)20} = 74.08182$, whence you can confirm that Mr. Pay and Mrs. Free have the exact same amount of money after 20 years even though Mr. Pay started off with much more. This is precisely why it is justifiable to say that the present value of 150 basis points for the next 20 years is $(100 - 74.08) = 25.92\%$, regardless of interest rates or growth rates.

11.3 A Simple Guaranteed Minimum Accumulation Benefit

Returning now to the topic of guarantees inside variable annuities: one type of guarantee that is commonly selected when purchasing a VA is the guaranteed minimum accumulation benefit (GMAB). In a simple scenario, assume that today ($t = 0$) you invest \$1 in a variable annuity. The random market value of the VA at maturity is

$$S_T = S_0 e^{\tilde{g}T}, \quad (11.21)$$

where $\tilde{g}$ is the random annualized return during time $[0, T]$. The guaranteed minimum accumulation benefit is then

$$S_T^* = \max[S_T, S_0 e^{gT}] \quad (11.22)$$

$$= S_T + \max[0, S_0 e^{gT} - S_T], \quad (11.23)$$

where g is a guaranteed growth rate. The GMAB is equivalent to a long position in the asset supporting the VA plus a put option struck at the initial investment level plus the minimal interest guarantee. This put can be valued using the BSM formula from Section 11.1.

Table 11.3. *Annual fee (in basis points) needed to hedge the death benefit—Female*

Age	Type of death benefit ^a		
	Money-back	5% roll-up	Look-back
30	0.30	1.77	15.10
40	0.80	4.45	18.90
50	2.00	10.84	24.60
60	5.00	21.60	32.80
65	7.60	22.50	36.10

^a Based on maturity at $x = 75$ years.
Note: GoMa mortality based on Tables 3.6 and 3.7; $r = 6\%$, $\sigma = 20\%$.

11.4 The Guaranteed Minimum Death Benefit

Next we examine the principal protection offered to beneficiaries upon the policyholder’s death—namely, the guaranteed minimum death benefit (GMDB). In order to determine the value of this embedded option, I must first state that the no-arbitrage value of a put option that matures at a random time T_x is

$$E[E[\max[S_0e^{gT} - S_0e^{\tilde{g}T}, 0] \mid T]].$$
 (11.24)

This is equivalent to

$$\Omega_x = \int_0^\tau f_x(t) \text{BSP}(t) \, dt,$$
 (11.25)

where τ is the termination date of the VA death benefit guarantee. Note that $f_x(t)$ is the probability density function of the time of death and $\text{BSP}(t)$ is the Black–Scholes/Merton put formula. Finally, we locate a value of k in equation (11.18) such that $E[K_{T_x}] = \Omega_x$.

When purchasing an annuity contract, one would typically be faced with a choice of different death benefit types; Table 11.3 displays several numerical examples. A 60-year-old female who purchases an annuity contract is entitled to a basic “money-back” guaranteed death benefit, where all invested funds would be returned to her beneficiaries in the event of her death. Under the usual GoMa parameters from Chapter 3, the annual fee that should be paid to the company for offering this option is 5 basis points. Alternatively, she may choose an enhanced death benefit with a 5% “roll-up,” where the death benefit is guaranteed to consist of at least the original premium paid,

Table 11.4. Annual fee (in basis points) needed to hedge the death benefit—Male

Age	Type of death benefit ^a		
	Money-back	5% roll-up	Look-back
30	0.40	3.24	25.00
40	1.30	7.96	31.60
50	3.50	19.20	41.80
60	8.70	37.50	56.40
65	13.00	39.30	62.50

^a Based on maturity at $x = 75$ years.

Note: GoMa mortality based on Tables 3.6 and 3.7; $r = 6\%$, $\sigma = 20\%$.

grown by an annual rate of 5%. In this case, her fee would rise to 21.6 basis points. Finally, in this example she may actually prefer a “look-back” option, where the payout to her beneficiaries would depend on the highest account value of past contract “anniversaries.” This is the most valuable option, for which she must pay 32.8 basis points per year. Of course, the older the individual purchasing the annuity the more she must pay for the guarantee, since she has fewer remaining lifetime years to compensate the company for the option.

Table 11.4 displays comparable values for a male annuitant. Although our observations concerning the previous table still apply, mortality differences result in the male paying more than the female for the same guarantee.

11.5 Special Case: Exponential Mortality

For general mortality it is very difficult to obtain a closed-form solution for the GMDB option expressed in equation (11.25). However, when T_x is exponentially distributed (so that $f_x(t) = \lambda e^{-\lambda t}$), we have

$$\Omega_x = \frac{\lambda}{2(r - g + \lambda)} \left(1 - \frac{b_2}{\sqrt{b_2^2 + 2(r - g + \lambda)}} \right) - \frac{\lambda}{2(k + \lambda)} \left(1 - \frac{b_1}{\sqrt{b_1^2 + 2(k + \lambda)}} \right), \quad (11.26)$$

where

$$b_1 = \frac{r - g - k + \frac{1}{2}\sigma^2}{\sigma}, \quad b_2 = \frac{r - g - k - \frac{1}{2}\sigma^2}{\sigma}. \quad (11.27)$$

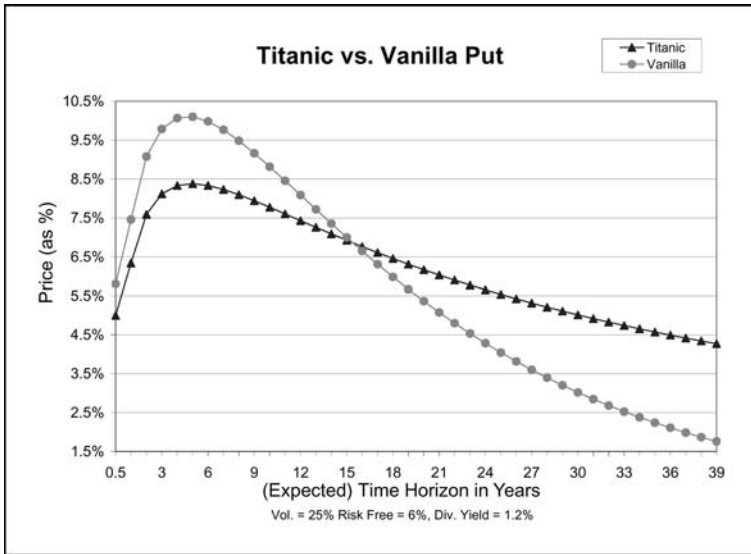


Figure 11.2

Equation (11.26) is the analogue of the BSM equation for the value of a put option, with the option's maturity now at a random (exponentially distributed) time T instead of at a fixed (deterministic) time T . The equation itself involves all the usual suspects from option pricing: the risk-free (valuation) rate r , the underlying security's investment volatility σ , the growth rate of the strike price g , the continuous insurance fees k , and the instantaneous mortality rate λ . The constants b_1 and b_2 are used as shorthand notation to simplify the main formula. Note that the subscript x in Ω_x indicates that the Titanic put option value is dependent on the purchaser's age—though only to the extent that the instantaneous force of mortality λ is higher or lower.

To better understand the intuition and properties of the Titanic option value Ω_x under exponential mortality—as defined by equation (11.26)—I offer Figure 11.2. It displays the price of a Titanic option that expires or matures at a random time T_x as compared to a generic European-style put option that matures at a deterministic time T .

Both options give the holder the right (but not the obligation) to sell an underlying security—that is currently trading for \$100—at a fixed price of \$100. This is an at-the-money put option in a 25% volatility environment, with a dividend yield (insurance fee) of 1.2% and a valuation rate of 6%. Using the symbols in equation (11.16), we have $\sigma = 0.25$, $r = 0.06$, $k = 0.012$, and $g = 0$, since the strike price does not change over time.

The law of mortality governing the Titanic put option gives us $\Pr[T_x > t] = e^{-\lambda t}$, which implies that $E[T_x] = 1/\lambda$. In the same figure, the European-style put option is constructed so that $T = 1/\lambda$ as well. In other words, the Titanic option *is expected to* mature at $1/\lambda$ whereas the European-style (vanilla) put option *will* mature at time $T = 1/\lambda$. For example: a European-style put option that matures in 3 years is worth approximately 9.7% of the underlying security value, whereas a Titanic put option—which is expected to mature in $E[T_x] = 3$ years under a $\lambda = 1/3$ mortality rate—is worth only 8.11% on the underlying security. As the graph illustrates, the European-style option is worth more. In fact, when $T = 5$ years, the European-style put is worth approximately 10% of the underlying account value whereas the Titanic option (with mortality rate $\lambda = 1/5$) is worth only 8.3%. Note the hump-shaped structure of the option value as a function of T (and of $E[T_x]$). As the (expected) maturity horizon increases from zero to approximately four years, the option value increases. Then, as the horizon increases, both option values decline toward zero. The European put's value declines more rapidly than the Titanic put's.

The intuition for this result is as follows. First, a European-style put option that promises only to return your \$100 at maturity should not be worth much if that maturity is 30 or 40 years from now. In contrast, the Titanic option can be viewed as a weighted average—see equation (11.15)—of the generic option value. The weighting is exponential, so some (small) weight is attached to the long horizon and some (larger) weight is attached to the short horizon. Add all these weights up and you should get the total picture. At a crude approximation, a Titanic option that matures in 5 years on average is the weighted average of a European option that matures in 1, 2, 3, ... years.

If $r = g$ and $k = 0$, then the basic formula in (11.26) further collapses to

$$\Omega_x = \frac{1}{\sqrt{1 + 8\lambda/\sigma^2}}. \quad (11.28)$$

This represents the case where the risk-free valuation rate is precisely equal to the rate at which the strike price is increasing over time. Thus, if $r = 5\%$ then the Titanic put option is assumed to have a strike price of \$1 at time zero, of $\$1e^{0.05}$ at time $t = 1$, of $\$1e^{(0.05)^2}$ at time $t = 2$, and so on. The owner of a Titanic put option can sell (put) the investments back to the insurance company at progressively higher values, which are determined by the valuation rate r . Guaranteeing a risk-free return might sound too good to be true, but you must wait until death to “cash in.” And, if $\lambda = 0$ and the holder never dies, $\Omega_x = 1$.

Table 11.5. *Value of exponential Titanic option*

Mortality	Volatility σ					
	5%	10%	15%	20%	25%	30%
$\lambda = 2\%$	12.40%	24.25%	35.11%	44.72%	53.00%	60.00%
$\lambda = 4\%$	8.80%	17.41%	25.63%	33.33%	40.42%	46.85%
$\lambda = 6\%$	7.20%	14.29%	21.16%	27.74%	33.94%	39.74%
$\lambda = 8\%$	6.24%	12.40%	18.43%	24.25%	29.83%	35.11%
$\lambda = 10\%$	5.58%	11.11%	16.54%	21.82%	26.92%	31.80%
$\lambda = 12\%$	5.10%	10.15%	15.13%	20.00%	24.72%	29.28%
$\lambda = 14\%$	4.72%	9.41%	14.03%	18.57%	22.99%	27.27%
$\lambda = 16\%$	4.42%	8.80%	13.14%	17.41%	21.58%	25.63%

Notes: $r = g, k = 0$. Value given as percentage of initial account value.

Using equation (11.28), Table 11.5 illustrates how the value of the exponential Titanic option is affected by varying values of λ and σ . As you can see, when λ is held constant, the value of the option increases with a growing volatility. Conversely, for any constant σ , the option decreases in value as λ increases. Thus, for instance, an x -year-old individual with an expected remaining lifetime of $E[T] = 10$ (which implies a $\lambda = 10\%$ mortality rate) who invests in a collection of mutual funds with a risk-free rate-of-return guarantee at death has an option that is “worth” \$21.82 for each \$100 investment. This assumes a portfolio volatility of $\sigma = 20\%$, which is reasonable for equity-based portfolios.

11.6 The Guaranteed Minimum Withdrawal Benefit

Another type of guarantee that has become very popular lately is the guaranteed minimum withdrawal benefit (GMWB), which is available as a “rider” to a VA policy. The GMWB rider promises to pay an annual dividend of G dollars per \$100 of original investment, regardless of how the actual account performs. (Recall that the investments within the variable annuity account can move up and down depending on actual market performance.) Furthermore, this dividend or payment of G per year will continue until the entire \$100 has been returned to the policyholder. So even if the VA account value collapses from \$100 to nothing (say) one day after the policy is acquired, the insurance company guarantees to continue making payments of G per year until the \$100 has been “paid back.” Of course, as with any other VA policy, the contract holder can surrender or lapse the policy and receive the market value (minus any surrender charges).

Here is the mathematics of how to understand this unique guarantee. Let W_t denote the market value of the underlying variable annuity at any future time $t \geq 0$, with an arbitrary (but innocuous) assumption that $W_0 = 100$ dollars. Under a typical GMWB structure, the policyholder is guaranteed to be able to withdraw at most $G = 7$ dollars per annum. The guarantee remains in effect until the entire \$100 has been disbursed, which is a period of at least $100/7 = 14.28$ years. Even in the extreme scenario where the initial $W_0 = 100$ collapses to a zero value one day after the policy is purchased, the investor will be made whole—albeit over an extended period of 14.28 years. Of course, in any year the policyholder may withdraw an amount of less than $G = 7$ dollars, which would extend the life of the guarantee; conversely, withdrawing an amount greater than $G = 7$ dollars would reduce the value *and* life of the guarantee. I shall proceed by assuming the policyholder withdraws no more and no less than the $G = 7$ dollars per annum; this is called the *passive* or *static* approach. Most (if not all) insurance companies assume this type of behavior on the part of policyholders.

Following the same setup as before, I assume that the actual dynamics of the assets underlying the VA policy (i.e., prior to deduction of any insurance fees) obeys the basic stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dB_t. \quad (11.29)$$

The value W_t of the VA subaccount incorporates two additional effects: proportional insurance fees and withdrawals. The account value satisfies

$$dW_t = (\mu - k)W_t dt - G dt + \sigma W_t dB_t, \quad (11.30)$$

at least while $W_t > 0$. If the account value W_t ever reaches zero, it remains there. That is: equation (11.30) holds for $t < \tau_0$, and $W_t = 0$ for $t \geq \tau_0$.

The solution to the SDE (11.30) can be written as

$$W_T = \exp\left\{(\mu - k - \tfrac{1}{2}\sigma^2)T + \sigma B_T\right\} \\ \times \max\left[0, \left(W_0 - G \int_0^T \exp\left\{-(\mu - k - \tfrac{1}{2}\sigma^2)t - \sigma B_t\right\} dt\right)\right]. \quad (11.31)$$

The first thing to note about the dynamics in equations (11.30) and (11.31) is that—since $G > 0$, which means that the process includes forced consumption of some dollar(s)—the value of W_t can in fact hit zero at some point $t > 0$. As soon as the integral term in (11.31) exceeds W_0/G , the quantity within brackets will become negative. This is in contrast to a standard

geometric Brownian motion, which is the term multiplying the bracketed portion of (11.31) that can never hit zero in finite time. The guaranteed ability to withdraw G per annum until time $T = W_0/G$ is of value *if and only if* the process W_t hits zero prior to T . Indeed, for those sample paths for which the ruin time occurs after T , the insurance option has a zero payout because the minimum withdrawal would have been satisfied endogenously, even without an explicit guarantee provided by the insurance company.

Given the importance of the ruin time in the classification and understanding of this financial guarantee, here I introduce an expression for the probability of ruin of the process W_t within the time period $[0, t]$:

$$\begin{aligned}\Pr\left[\inf_{0 \leq s \leq t} W_s = 0\right] &= \Pr\left[\int_0^t \exp\left\{-(\mu - \alpha - \tfrac{1}{2}\sigma^2)s - \sigma B_s\right\} ds \geq \frac{W_0}{G}\right] \\ &= \Pr\left[X_t \geq \frac{W_0}{G}\right],\end{aligned}\tag{11.32}$$

where the new term X_t is defined as equal to the integral in the middle of (11.32). Note the analogy between the probability of ruin and the integral of the (inverse) of the geometric Brownian motion. This is yet another manifestation of the stochastic present value, which was at the core of our approximations in Chapter 9. As mentioned in the appendix of that chapter, the seemingly counterintuitive relationship between the infimum of a process and the integral of an exponential Brownian motion follows because equation (11.31) cannot reach zero until the integral X_t exceeds W_0/G . Note also that X_t is monotonically increasing in t . Thus, once X_t exceeds W_0/G we have $W_t = 0$, and it can never recover and go back above zero. It is quite easy to demonstrate that the probability of ruin is increasing in the withdrawal rate G and likewise that, the greater the time t , the higher the probability of ruin.

Assume the arithmetic average return (after money management fees but before insurance guarantee fees) is expected to be $\mu = 9\%$ per annum with a historical market volatility of $\sigma = 18\%$. Also, we let the insurance fee for this particular GMWB rider be set to $k = 0.40\%$ per annum, which is consistent with current market pricing for these products. In this case, while $W_t > 0$ the parameterized dynamics of the investment is

$$dW_t = ((0.086)W_t - 7)dt + 0.18W_t dB_t, \quad W_0 = 100. \tag{11.33}$$

Using numerical methods to obtain the ruin probability during the first $T = 14.28$ years yields a ruin probability of 11.7% (see Section 11.7 for references). In other words, there is approximately an 88.3% chance that the

Table 11.6. *GMWB payoff and the probability of ruin within 14.28 years*

Volatility σ	Expected return μ of subaccounts				
	4%	6%	8%	10%	12%
10%	19.0%	7.0%	1.7%	0.3%	0.04%
15%	31.4%	18.5%	9.3%	4.1%	1.60%
18%	37.8%	25.5%	15.5%	8.6%	4.40%
25%	49.9%	39.6%	30.5%	22.2%	15.50%

Note: $k = 40$ -basis-point fee.

policy will survive to the end of the guaranteed horizon even if the policyholder withdraws the maximum allowable amount each year. But if we increase the investment return volatility to $\sigma = 25\%$ per annum, the ruin probability increases to 26.2%. If we reduce the expected (arithmetic average) return to $\mu = 6\%$ and maintain a high $\sigma = 25\%$ volatility then the probability of ruin increases to 39.9%; these are clearly nontrivial amounts. Table 11.6 displays the probabilities under various risk–return combinations.

Observe that, if the expected investment return is increased to $\mu = 12\%$ with a volatility of $\sigma = 10\%$, then the probability that the withdrawals of $G = 7$ dollars per annum will actually exhaust (ruin) the policy prior to time $T = 14.28$ is less than half of a percent. Thus, an overly optimistic insurance actuary focused on real-world payout probabilities risks ignoring this event altogether.

In any case, the probability of ex ante guarantee usage ranges from 0.5% to 50%, depending on our assumptions about asset characteristics and returns, and these usage probabilities will affect the setting of traditional insurance reserves. The relevant question to a financial economist interested in the fair value of liabilities is: How much does it cost the insurance company to hedge this guarantee in the capital market?

I now illustrate how to bifurcate the product into a collection of strip bonds (or a term-certain annuity) and a complex option in the form of a so-called Quanto Asian put (QAP). Note that by definition $T = W_0/G$ (since the product terminates or matures when all the funds have been returned), so (11.31) can be written as

$$W_T = W_0 \exp\left\{\left(\mu - k - \frac{1}{2}\sigma^2\right)T + \sigma B_T\right\} \\ \times \max\left[0, \left(1 - \frac{1}{T} \int_0^T \exp\left\{-\left(\mu - k - \frac{1}{2}\sigma^2\right)s - \sigma B_s\right\} ds\right)\right];$$

here the

$$[\text{QAP option payoff}] := W_T, \quad (11.34)$$

since the holder of the variable annuity policy is guaranteed to receive any remaining funds in the account at time $T = W_0/G$. Remember that the policyholder is also entitled to the periodic income flow in addition to the (possibly zero) maturity value of the account. The maturity value of the *periodic income* is

$$G \int_0^T e^{rt} dt = \frac{G}{r} (e^{rT} - 1). \quad (11.35)$$

The no-arbitrage, time-0 present value of the GMWB cash-flow package is therefore

$$e^{-rT} E^*[W_T] + \frac{G}{r} (1 - e^{-rT}), \quad (11.36)$$

where $E^*[\cdot]$ denotes the expectation under the option pricing measure, for which the real-world drift μ is replaced by the risk-free rate r .

Finally, for the GMWB to be fairly priced, at inception we must have that the amount invested in the product W_0 is equal to the value of the cash-flow package, where $T = W_0/G$:

$$W_0 = e^{-rT} E^*[W_T] + \frac{G}{r} (1 - e^{-rT}). \quad (11.37)$$

Equation (11.37) is one of our main results. It states that, for the product to be fairly structured, the initial purchase price must equal the cost of the term-certain annuity plus the exotic option. For any given (r, σ) pair we can locate the (k, G) curve across which the product is fairly priced, and this implies the equality of (11.37).

I further claim that the option component is effectively a Quanto Asian put defined on the inverse of the account price process. To see this, define a new (reciprocal) process as follows:

$$Y_t = S_t^{-1} = \exp\left\{-\left(r - k - \frac{1}{2}\sigma^2\right)t - \sigma B_t\right\}, \quad Y_0 = 1. \quad (11.38)$$

One can think of Y_t as the number of VA subaccount units that one dollar can buy, similar to the number of euros or yen that one dollar can purchase in the currency market. The inverse, $S_t = Y_t^{-1}$, is the value of one VA subaccount unit in dollars, analogous to the price of one euro or one yen in U.S. dollars. Now let

$$A_T := \frac{1}{T} \int_0^T Y_t dt, \quad (11.39)$$

which is an average of the reciprocal account value. The payoff from this option at maturity can now be written as

$$[\text{QAP option payoff}] := W_0 \frac{\max[1 - A_T, 0]}{Y_T}. \quad (11.40)$$

This represents W_0 units of a Quanto (fixed-strike) Asian put option. In sum, scaling everything by the initial premium, a fairly priced product *at inception* implies the relationship

$$e^{-rT} E^* \left[\frac{\max[1 - A_T, 0]}{Y_T} \right] + \frac{G}{r} (1 - e^{-rT}) = 1. \quad (11.41)$$

Given values of the other parameters, the fair insurance fee k can be obtained by solving this equation.

Thus, our main qualitative insight is that, under a static perspective, this product can be decomposed into the following items:

1. a term-certain annuity paying G per annum for a period of $T = W_0/G$ years; plus
2. a Quanto Asian put on the aforementioned reciprocal variable annuity account.

For example: with an initial deposit of $W_0 = \$100$, a guaranteed withdrawal amount of $G = 7$ dollars per annum, and an interest rate of $r = 0.06$, the time-0 cost of the term-certain annuity component is \$67.15. The remaining \$32.85 would be used to purchase the option, and k is determined so that this represents the fair option value. One can think of a VA with a GMWB as consisting of 67% term-certain annuity and 32% QAP option. In contrast, at a (lower) interest rate of $r = 0.05$, the cost of the term-certain annuity would be (a higher) \$71.46 and only \$28.54 would be used to purchase the required option.

Table 11.7 displays the required insurance fee that would lead to equality in equation (11.36) or (11.41) under a number of different volatility values. Note the fixed-point nature of the problem. Once the volatility σ , interest rate r , and guarantee rate G have been selected, we must numerically search for a fee value k that yields equality in (11.41). We price the QAP option using techniques referenced in Section 11.7. For example, if the VA guarantees a 7% withdrawal and if the pricing volatility is $\sigma = 20\%$, then the fair insurance fee would be approximately $k = 73$ basis points of assets per annum. Stated differently, a financial package that includes a stream of \$7-per-annum income (in continuous time) plus a Quanto Asian put that

Table 11.7. *Impact of GMWB rate and subaccount volatility on required fee k*

Guarantee rate W_0/G	Maturity (years) $T = 1/g$	Investment volatility	
		$\sigma = 20\%$	$\sigma = 30\%$
4%	25.00	23 b.p.	60 b.p.
5%	20.00	37 b.p.	90 b.p.
6%	16.67	54 b.p.	123 b.p.
7%	14.29	73 b.p.	158 b.p.
8%	12.50	94 b.p.	194 b.p.
9%	11.11	117 b.p.	232 b.p.
10%	10.00	140 b.p.	271 b.p.
15%	6.67	272 b.p.	475 b.p.

Notes: b.p. = basis points. Assumes valuation rate of $r = 5\%$.

matures in exactly $T = 14.29$ years is a package worth precisely $W_0 = 100$ when the investment on which the option is struck is “leaking” a dividend yield of 73 basis points per annum. If the guarantee is reduced to $G = 4\%$ —which implies that the product matures in $T = 25$ years—then the fair insurance fee is only 23 basis points. Likewise, if the guarantee is increased to $W_0/G = 9\%$ —which implies that the product matures in $T = 11.11$ years—then the fair insurance fee is 117 basis points. The most common GMWB guarantee being offered on variable annuities is $W_0/G = 7\%$, which even under a conservative $\sigma = 15\%$ volatility implies an insurance fee of 40 basis points.

Our equating of the GMWB with a term-certain annuity plus a QAP is useful from several points of view. First of all, though I used numerical techniques to value the embedded option, this is not essential because there are a variety of other well-studied approaches to the valuation of Asian options. Second, there is an established over-the-counter market for Asian options, which raises the possibility of hedging via these products (instead of dynamic hedging). Finally, there is a body of practical experience with the hedging of Asian options, which means that the QAP and hence the GMWB are both more familiar products than they may first appear.

11.7 Further Reading

I have examined some of the derivative securities within deferred variable annuities that are associated with pension annuity policies in the United

States. The objective of this chapter was to give the reader a flavor of the types of models and issues being explored by researchers in the field, not to provide a definitive guide or reference on the embedded options. The material in this chapter draws heavily from Milevsky and Posner (2001) and from Milevsky and Salisbury (2006). Additional recommended sources on the topic of guaranteed annuity options include Ballotta and Haberman (2003) and Boyle and Hardy (2003) as well as Milevsky and Promislow (2001). The numerical solution to the ruin probability of the GMWB and the pricing of the QAP are described in Huang et al. (2004). Of course, the “bible” on derivative pricing remains the classic book by John Hull (2002); for those readers who want to “learn” option pricing, there is no better place (at least in my opinion). In their 1976 paper, Brennan and Schwartz were the first to conduct a rigorous treatment of options inside insurance contracts, which is the basis for (11.24).

In the last ten years there has been an explosion of scholarly research and academic papers that have focused on the financial options embedded within pension plans and insurance policies. It is, of course, impossible to do justice to each citable author in the field, and clearly this chapter has been biased by my own interests and research work. However, if you are interested in learning much more about pricing these types of options, I recommend *Investment Guarantees* (Hardy 2003); that book contains a much more detailed examination of alternative models to the simple Brownian motion framework used in this chapter. The merging of mortality and investment derivatives is just getting started.

11.8 Notation

Ω_x —value of a Titanic put option that promises a money-back guarantee at death, where the guarantee level (strike price) increases each year by $g\%$

$BSM(t)$ —value of a generic put option that promises a money-back guarantee at time t

k —instantaneous fee that is paid for the insurance guarantee

K —discounted value of the fee that is paid for the insurance